

Exact randomization inference with uncertain record identities

Gaurav Sood

October 2026

Abstract

An experimental conclusion can depend on how outcome records are assigned to experimental units. We develop a sensitivity analysis that retains candidate links, one-to-one ownership, and a bound on changed accepted identities. Under complete randomization and full overlap, every feasible outcome permutation has the same sharp-null reference distribution. The largest Fisher p-value therefore follows from the smallest attainable absolute contrast; the smallest p-value follows from the linear contrast endpoints. Integer optimization returns the linkages attaining both extremes. We also report the minimum identity-change budget that removes rejection and give a closed-form binary comparator that uses accuracy information without candidate restrictions. Exhaustive checks establish agreement with direct enumeration, and an identifier-based simulation measures what the candidate graph adds. In these synthetic settings, gains are modest and substantial power loss remains. The guarantee holds when the true linkage belongs to the stated set.

1 The question and the contribution

A researcher has randomized treatment assignments in one file and outcomes in another. Several outcome records may plausibly belong to each experimental unit. The question is whether the sharp no-effect null can be rejected under every permitted assignment of records to units. A linkage accuracy rate restricts how many accepted identities may be wrong. Candidate links additionally restrict which outcomes can move between which units. These restrictions can change the answer even when the accepted matching and error budget are held fixed.

Our contribution is a specified randomization sensitivity procedure over record identities. It computes the exact p-value endpoints for a stated graph and error budget, returns the attaining identities, and identifies the budget at which rejection first fails. Its useful specialization is the combination of the experimental assignment law with conservation of whole donor records. A corrected outcome vector that cannot arise from a feasible matching is excluded. The algorithm does not require probabilities for individual links.

Optimization of statistics over matching confidence sets is established [1]; their permutation test concerns a hypothesized matching, whereas here the randomness is the experimental assignment. Integer-programming sensitivity to binary outcome misclassification is also established [2]; here the budget counts changed identities rather than altered outcomes, and whole donor records are conserved under candidate restrictions. Permutation-set inference under a Gaussian regression model uses a different source of randomness [3]. The result here supplies the full-bijection Fisher reference law and its reduction to attainable contrasts, using standard assignment constraints, absolute-value optimization, and confidence-set maximization. Linkage-induced causal bias and loss of power motivate a substantial literature [5, 6].

2 Experimental units and feasible identities

Fix N units with potential outcomes $Y_i(0), Y_i(1)$. Exactly m receive treatment, uniformly over the $\binom{N}{m}$ assignments; write $c = N - m$. Assignment Z is observed without error. The donor file contains exactly one realized outcome for every unit and no other outcomes. Its rows $y_1, \dots, y_N$ have unknown true ownership π^* , a bijection. Thus $Y_i = Y_i(Z_i) = y_{\pi^*(i)}$.

Let G be a bipartite candidate graph and π_0 an accepted bijection. For integer K , define

$$\mathcal{M}(G, K) = \left\{ \pi : \pi \text{ bijective, } (i, \pi(i)) \in G \ \forall i, \quad \sum_i 1\{\pi(i) \neq \pi_0(i)\} \leq K \right\}.$$

The analysis requires a nonempty set. Unless stated otherwise, it assumes $\pi^* \in \mathcal{M}(G, K)$. A donor supplies its complete record: treating each field as independently reassignable would describe a different uncertainty set. The present test concerns known assignment and uncertain outcomes; it is not a general correction for jointly mislinked treatment, covariates, and outcomes.

For any feasible π , the observed contrast is

$$T(\pi) = \sum_i w_i y_{\pi(i)}, \quad w_i = \frac{Z_i}{m} - \frac{1 - Z_i}{c}.$$

Let $L = \min_{\pi \in \mathcal{M}} T(\pi)$ and $U = \max_{\pi \in \mathcal{M}} T(\pi)$. These are extrema of reconstructed estimates. They are not confidence limits for the finite-population average effect $N^{-1} \sum_i \{Y_i(1) - Y_i(0)\}$.

3 Exact p-value endpoints

We test Fisher's sharp null $Y_i(1) = Y_i(0)$ for every unit. For a uniform size- m assignment A , define the common tail function

$$H(s) = \Pr_A \left(\left| \sum_i \left\{ \frac{A_i}{m} - \frac{1 - A_i}{c} \right\} y_i \right| \geq s \right).$$

Proposition 1 (Common reference law). *Under complete randomization, the distribution of the absolute contrast is identical for every full permutation of the donor outcomes. Consequently, if $d = \min_{\pi \in \mathcal{M}} |T(\pi)|$ and $D = \max(|L|, |U|)$, then*

$$p_{\max} = \max_{\pi \in \mathcal{M}} p(\pi) = H(d), \quad p_{\min} = \min_{\pi \in \mathcal{M}} p(\pi) = H(D).$$

Both endpoints are attained by feasible linkages.

Proof. A permutation of unit labels maps the collection of size- m treatment subsets bijectively onto itself. Each subset has the same probability, so permuting the outcome vector preserves its randomization law. The function H is nonincreasing, and the nonempty finite feasible set attains the minimum and maximum absolute contrasts. The maximum absolute contrast is attained at a linear endpoint. $\square$

The maximum p-value answers whether every permitted linkage rejects. The minimum p-value shows how favorable a permitted reconstruction could be. Selecting the linkage with the smallest p-value after observing outcomes does not yield a valid robust test. The reference law uses the full experimental assignment support; it does not condition assignment on the observed candidate graph.

For binary outcomes with s_y successes, treated successes follow $Q \sim \text{Hypergeometric}(N, s_y, m)$, and

$$H(s) = \sum_{q=\max(0, m-N+s_y)}^{\min(m, s_y)} \frac{\binom{s_y}{q} \binom{N-s_y}{m-q}}{\binom{N}{m}} \mathbf{1} \left\{ \left| \frac{q}{m} - \frac{s_y - q}{c} \right| \geq s \right\}.$$

Any two-valued outcome has the same calculation after affine recoding. For general outcomes, the implementation enumerates the assignment distribution subject to an explicit size limit. It optimizes over linkages without enumerating them. This does not imply that integer optimization is inexpensive for arbitrary graphs or outcomes.

3.1 Inference when the graph can miss truth

Proposition 2 (Containment-adjusted validity). *Suppose $\Pr\{\pi^* \notin \mathcal{M}(G, K)\} \leq \delta$ under every null population under consideration. Then $p_{\text{rob}} = \min(1, p_{\text{max}} + \delta)$ satisfies $\Pr(p_{\text{rob}} \leq \alpha) \leq \alpha$.*

Proof. On containment, $p_{\text{max}} \geq p(\pi^*)$ pointwise. For $\delta \leq \alpha < 1$, rejection therefore implies either containment failure or $p(\pi^*) \leq \alpha - \delta$. The first event has probability at most δ and the second at most $\alpha - \delta$ by oracle randomization validity. If $\alpha < \delta$, rejection is impossible; $\alpha = 1$ is immediate. $\square$

This is the established confidence-set maximization argument [4]. The graph may be data dependent; independence between graph construction and outcomes is not required by the proof. What is required is the joint containment bound under the relevant repeated experiment. Pairwise candidate recall is not the probability that the entire true matching survives. The software accepts δ ; it does not estimate or certify it.

4 Optimization and an identity breakdown budget

Use binary variables x_{ij} for allowed edges. Require each row and column to sum to one and $N - \sum_i x_{i, \pi_0(i)} \leq K$. Minimizing and maximizing $\sum_{ij} w_i y_j x_{ij}$ gives L, U . For d , add $t \geq 0$ and minimize t subject to

$$-t \leq \sum_{ij} w_i y_j x_{ij} \leq t.$$

For two-valued outcomes, we optimize the equivalent integer objective $|Nq - ms_y|$ and bound it below by its minimum over hypergeometric support. This prevents a search for zero when the outcome lattice excludes it. The returned assignments are checked for feasibility and agreement with the objectives. An interval $[L, U]$ spanning zero does not guarantee an attainable zero. For example, the only contrasts might be -2 and 2 . Substituting the distance of $[L, U]$ from zero would discard the discreteness that the integer program preserves.

An identity swap between anchors i, k changes the contrast by $(w_i - w_k)(y_{\pi(k)} - y_{\pi(i)})$. Within-arm swaps preserve the unadjusted contrast; cross-arm swaps matter according to the outcome difference. Arbitrary matching differences decompose into alternating cycles. This explains why error counts and candidate counts alone do not determine uncertainty. These are standard assignment properties applied to the causal contrast, consistent with earlier observations about harmless within-subclass linkage errors [5].

For breakdown analysis require the accepted map itself to satisfy the graph constraints, so the zero-budget set is nonempty. Define the identity breakdown budget by

$$K_{\text{break}} = \min\{K : p_{\text{rob}}(G, K) > \alpha\},$$

where rejection uses $p \leq \alpha$. The output is zero if the containment-adjusted accepted p-value already fails to reject and infinity if every graph-feasible map rejects. Because feasible sets grow with K , p_{rob} is nondecreasing, so binary search finds the first budget. It returns a linkage demonstrating the loss of rejection. This is an identity-constrained counterpart of warning accuracy for outcome misclassification [2].

4.1 A comparator that conserves donors

The accuracy-only comparison retains the same y, π_0, K and one-to-one constraint, but permits every anchor–donor edge. It does not invent replacement outcomes.

Proposition 3 (Binary accuracy-only support). *For binary outcomes, let q_0 be accepted treated successes and s_y total successes. Under the complete graph and at most K changed identities, attainable treated-success counts are exactly the integers in*

$$[\max\{0, m - N + s_y, q_0 - \lfloor K/2 \rfloor\}, \min\{m, s_y, q_0 + \lfloor K/2 \rfloor\}].$$

Proof. Changing treated successes by h requires at least $|h|$ changes of treated outcomes and $|h|$ opposite changes of control outcomes. Hence at least $2|h|$ identities must change. Disjoint swaps of opposite-valued donors across arms attain this lower bound for every count within the natural support. $\square$

Evaluating H at these counts gives an exact comparator without a complete-graph integer program. Arbitrary candidate graphs can leave gaps in attainable counts; the interval formula is not transferred to them. Equal-valued donors also cannot be collapsed if their allowed neighbors or accepted-identity costs differ.

5 Validation and the information gained from candidates

The code is checked against exhaustive linkages and assignments in small problems, including unequal arms, off-graph budgets, ties, disconnected components, and shared identity budgets. Tests compare p-value endpoints and their witnesses, rather than only solver objective values. Further checks establish scale and location invariance and compare breakdown budgets with exhaustive search. Outcomes are centered and scaled before optimization and tail comparisons, then reported contrasts are returned in their original units. This prevents an absolute numerical tolerance from making small-unit outcomes appear identically zero.

5.1 A controlled comparison

A deliberately constructed example keeps the accepted map, its identity accuracy, and the number of candidates per record fixed while changing which cross-arm exchanges are possible. Table 1 shows different robust conclusions. These are controlled information sets, not estimated graphs.

Enumerating every assignment in the accompanying design checks size and power without Monte Carlo uncertainty. Table 2 retains the cases where the graph gains little or substantial power is still lost. A narrow estimate range by itself does not remove experimental uncertainty.

Table 1: Same accepted identities and accuracy information, different candidate graphs. The controlled example fixes eight units, four treated, and an at-most-two identity-error budget. Both restricted graphs offer two candidates per unit. P-values are exact two-sided Fisher probabilities.

Information	Lower contrast	Upper contrast	Maximum p
Accuracy only	0.50	1.00	0.486
Within-arm candidates	1.00	1.00	0.029
Cross-arm candidates	0.50	1.00	0.486

Table 2: Exact rejection probabilities in the eight-unit design at the 5% level. All 70 assignments are enumerated, so there is no Monte Carlo error. The zero-effect row measures sharp-null size; other rows measure power under constant additive effects. Each method uses the same accepted linkage and error budget.

Effect	Oracle	Accuracy only	First graph	Second graph
0	0.029	0.000	0.029	0.000
1	0.014	0.000	0.014	0.000
2	1.000	0.000	0.086	0.086
4	1.000	0.000	0.086	0.086

5.2 Candidates generated from identifiers

The second comparison contains 960 evaluations and generates identifiers before treatment and outcomes. Records share a known geographic group and a short, potentially duplicated identifier. At most one identifier character is corrupted; candidate links require the same group and at most one character difference. True-link containment follows from this stated error support, without adding missing true edges after graph construction. A minimum-distance one-to-one matching, with random outcome-independent tie breaking, supplies the accepted map. Donor rows are shuffled so that row position does not reveal identity. There are ten geographic groups in each population; increasing sample size also increases the records per group and candidate ambiguity. This is not a scaling experiment that holds linkage difficulty fixed.

The graph and accuracy-only procedures receive the same accepted map and the same *oracle-known* number of incorrect accepted identities. This isolates the value of candidate restrictions; it is not a method for learning an accuracy budget from unvalidated records. Simulations cross two sample sizes, two corruption probabilities, uniform versus geographically varying baseline risks, and a sharp null plus two positive treatment-response increments. Complete randomization is independent of identifiers. Positive increments need not equal the average treatment effect because success probabilities are capped at one. Every scenario is retained, including weak gains and failures. A solver time limit gives a conservative nonrejection on failed optimizations; those runs are recorded as unresolved, not exact p-value endpoints. The protocol records the previously inspected controlled examples and separates timing pilots from final simulation seeds.

Table 3: Identifier-generated candidates: rejection frequencies and paired gains. Each row uses 40 replications at the 5% level. Corruption changes at most one identifier character; geographic baseline risks vary by the known identifier group. Each comparison uses the oracle-known accepted identity-error count. Rejection rates are proportions; the last column is graph minus accuracy-only rejection, with Monte Carlo standard error. Unresolved optimizations conservatively do not reject; failures are reported in the text. The effect column is a probability increment before capping.

N	Noise	Baseline	Effect	Oracle	Accuracy	Graph	Gain (SE)
40	0.25	Geographic	0.00	0.025	0.000	0.000	0.000 (0.000)
40	0.25	Geographic	0.25	0.175	0.000	0.025	0.025 (0.025)
40	0.25	Geographic	0.50	0.575	0.075	0.100	0.025 (0.025)
40	0.75	Geographic	0.00	0.000	0.000	0.000	0.000 (0.000)
40	0.75	Geographic	0.25	0.100	0.000	0.000	0.000 (0.000)
40	0.75	Geographic	0.50	0.575	0.000	0.100	0.100 (0.048)
40	0.25	Uniform	0.00	0.025	0.000	0.000	0.000 (0.000)
40	0.25	Uniform	0.25	0.225	0.050	0.050	0.000 (0.000)
40	0.25	Uniform	0.50	0.800	0.200	0.200	0.000 (0.000)
40	0.75	Uniform	0.00	0.025	0.000	0.000	0.000 (0.000)
40	0.75	Uniform	0.25	0.125	0.000	0.000	0.000 (0.000)
40	0.75	Uniform	0.50	0.850	0.025	0.050	0.025 (0.025)
80	0.25	Geographic	0.00	0.025	0.000	0.000	0.000 (0.000)
80	0.25	Geographic	0.25	0.500	0.000	0.000	0.000 (0.000)
80	0.25	Geographic	0.50	0.925	0.000	0.000	0.000 (0.000)
80	0.75	Geographic	0.00	0.075	0.000	0.000	0.000 (0.000)
80	0.75	Geographic	0.25	0.625	0.000	0.000	0.000 (0.000)
80	0.75	Geographic	0.50	0.950	0.000	0.000	0.000 (0.000)
80	0.25	Uniform	0.00	0.075	0.000	0.000	0.000 (0.000)
80	0.25	Uniform	0.25	0.500	0.000	0.000	0.000 (0.000)
80	0.25	Uniform	0.50	1.000	0.000	0.000	0.000 (0.000)
80	0.75	Uniform	0.00	0.025	0.000	0.000	0.000 (0.000)
80	0.75	Uniform	0.25	0.450	0.000	0.000	0.000 (0.000)
80	0.75	Uniform	0.50	0.975	0.000	0.000	0.000 (0.000)

At the larger probability increment, the graph procedure rejects in 18 of 320 evaluations, compared with 12 for accuracy alone and 266 with known identities. The gains are modest and substantial power loss remains. Across the positive-increment scenarios, the graph’s paired rejection gain ranges from 0.00 to 0.10. Mean candidate counts range from 2.3 to 4.1 across scenarios; mean accepted identity accuracy ranges from 0.41 to 0.87. Scenario median graph-test runtimes range from 0.001 to 0.010 seconds; the largest scenario 95th-percentile runtime is 3.808 seconds. These timings describe the tested graphs and hardware, not worst-case complexity. Of 960 evaluations, 14 did not return certified optima under the per-solver time limit and received the conservative nonrejection fallback. The intervals and standard errors in this comparison concern Monte Carlo uncertainty across synthetic populations and assignments. A zero estimated Monte Carlo standard error means all sampled paired differences were zero; it does not establish a zero population difference. Completed individual tests use exact hypergeometric reference probabilities. Synthetic name errors with known support and an oracle error budget do not establish performance for actual administrative linkage.

6 Scope and empirical use

The method covers Fisher’s sharp null under complete randomization with known assignment, when both files contain exactly the same units once. Blocked assignment, partial overlap, extra donors, covariate-dependent weights, and linked treatment labels change the reference law and need their own derivations.

For an application, supply an independently justified candidate set and error budget, compute the robust p-value and attaining map, and inspect the identity changes that matter. If whole-set coverage is uncertain, report sensitivity to omissions or a supported containment adjustment. A small maximized p-value means that every identity reconstruction retained by the stated restrictions rejects the sharp no-effect null.

References

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A Structure of the feasible contrast

These results describe the feasible set $\mathcal{M}(G)$ of perfect matchings in G without an error budget, except where the budget is named.

Proposition 4 (Candidates refine accuracy). *For the same π_0 and K , $\mathcal{M}(G, K) \subseteq \mathcal{M}(G_c, K)$, where G_c is the complete bipartite graph. Hence $L_G \geq L_c$, $U_G \leq U_c$, and $p_{\max}(G) \leq p_{\max}(G_c)$.*

Proof. Every graph-feasible map satisfies the accuracy-only constraints. Minima, maxima and maximum p-values over a subset are bounded by those over the superset. $\square$

The inequalities can be strict or equalities, and graphs with equal degrees need not be ordered by inclusion.

Proposition 5 (Zero width). *Fix $\pi \in \mathcal{M}(G)$. For an alternating cycle exchanging the donors of distinct anchors $i_1, \dots, i_r$ through edges $(i_k, \pi(i_{k+1})) \in G$ (indices cyclic), let*

$$D_C = \sum_{k=1}^r w_{i_k} \{y_{\pi(i_{k+1})} - y_{\pi(i_k)}\}.$$

Then $L = U$ over $\mathcal{M}(G)$ if and only if $D_C = 0$ for every such cycle.

Proof. If the width is zero, every feasible cycle flip preserves the contrast. Conversely, the edges on which any second perfect matching differs from π decompose into disjoint alternating cycles, and the contrast difference is the sum of their D_C . $\square$

A shared error budget can forbid single cycle flips while permitting coordinated ones, so the characterization is for $\mathcal{M}(G)$.

Proposition 6 (Components). *Without a shared budget, $\mathcal{M}(G)$ is the Cartesian product of the perfect matchings of the connected components of G , and L , U and $U - L$ add across components. A component within one treatment arm, or whose donors share one outcome value, contributes zero width. If component b is complete bipartite with n_b anchors, t_b of them treated, and sorted donor outcomes $y_{b(1)} \leq \dots \leq y_{b(n_b)}$, its width is*

$$W_b = \left(\frac{1}{m} + \frac{1}{c} \right) \left\{ \sum_{r=n_b-t_b+1}^{n_b} y_{b(r)} - \sum_{r=1}^{t_b} y_{b(r)} \right\}.$$

Proof. A perfect matching never crosses components and uses every donor of its component once, so the feasible set factorizes and the objective is a sum. A one-arm component contributes its constant weight times a fixed donor total. In a complete component the contribution is $(1/m + 1/c)$ times the sum of the t_b donors given to treated anchors, minus the fixed donor total divided by c ; any t_b -subset is attainable, so the smallest and largest such sums give the extrema. $\square$